HALOKAZI

Web-Based CV Builder, Career & Job Opportunity Platform

STAGE 5 - SYSTEM MODELING

System Modeling Documentation

Use Case Diagrams • Activity Diagrams • Entity Relationship Diagram • Data Flow Diagrams

Prepared for Project Submission

1. System Modeling Overview

This document presents the system modeling work completed for the Halokazi project. It brings together the functional, process, data and database models prepared during Stage 5. The diagrams are to be inserted from the completed draw.io modeling file.

The document covers use case diagrams, activity diagrams, the Entity Relationship Diagram (ERD), and Data Flow Diagrams (DFD Level 0 and Level 1). The descriptions provide the modeling details.

1.1 Modeling Artifacts

- UC-01 Overall Halokazi Use Case Diagram
- UC-02 Job Seeker Use Case Diagram
- UC-03 Employer Use Case Diagram
- UC-04 Admin Use Case Diagram
- UC-05 Super Admin Use Case Diagram
- AD-01 Job Seeker Registration, Verification & CV Builder
- AD-02 Employer Registration & Verification
- AD-03 Job Seeker Application
- AD-04 Employer Application Review & Status Update
- AD-05 Vacancy Creation & Lifecycle
- ERD Entity Relationship Diagram
- DFD Level 0 Halokazi Context Diagram
- DFD Level 1 Major Processes and Data Stores

2. Use Case Models

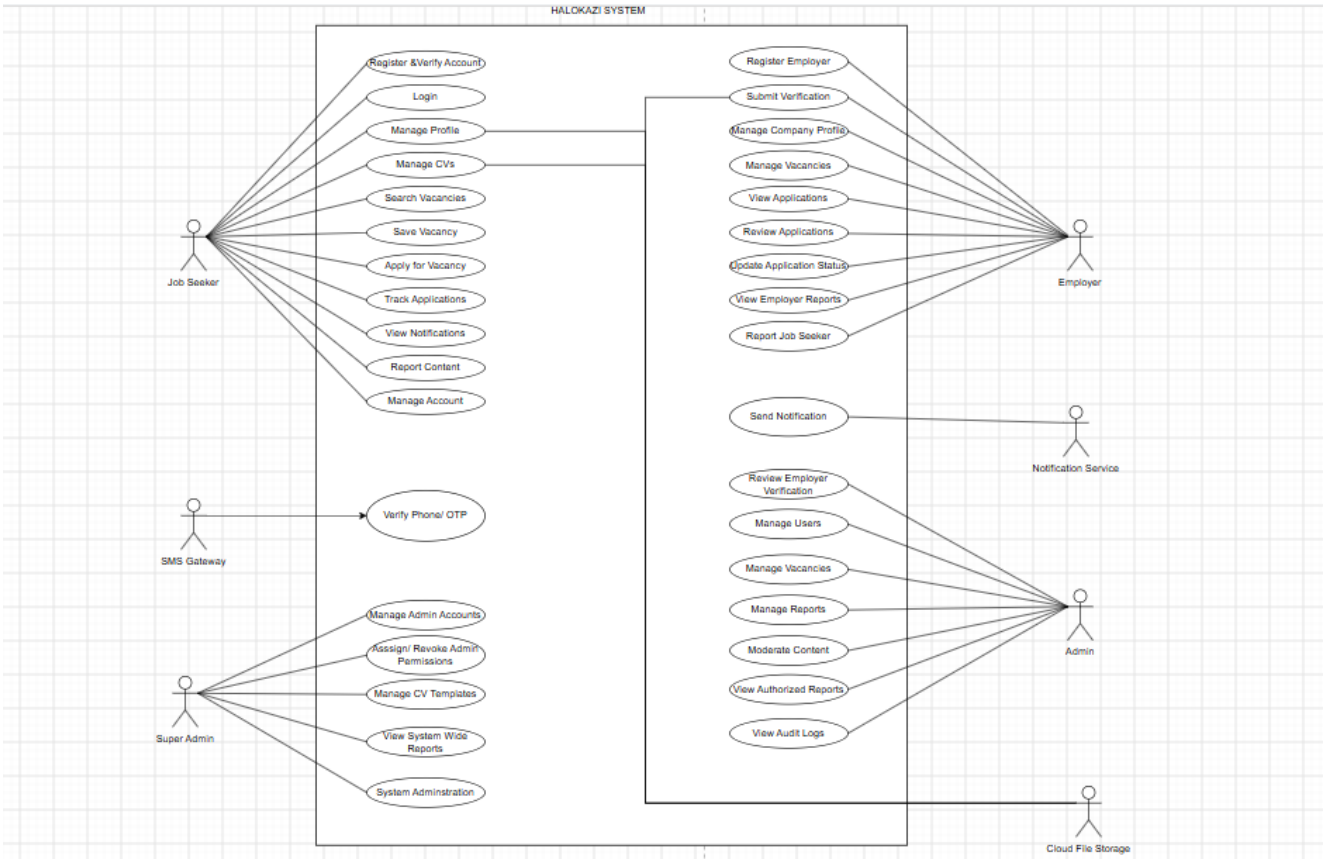
The use case models describe the main functions of Halokazi and the actors interacting with the platform.

2.1 Actors

Actor	Type	Main Interaction
Job Seeker	Primary	Registration, profile, CVs, vacancy search, applications, notifications and reporting.
Employer	Primary	Employer verification, company profile, vacancies, applications, reports and reporting Job Seekers.
Admin	Primary	Employer verification, user/vacancy management, reports, moderation and authorized administration.
Super Admin	Primary	Admin accounts, permissions, templates, system administration and system-wide reports.
SMS Gateway	Secondary	OTP/SMS delivery for phone verification.
Notification Service	Secondary	Application and relevant-job notifications.
Cloud File Storage	Secondary	CVs, profile files and employer verification documents.

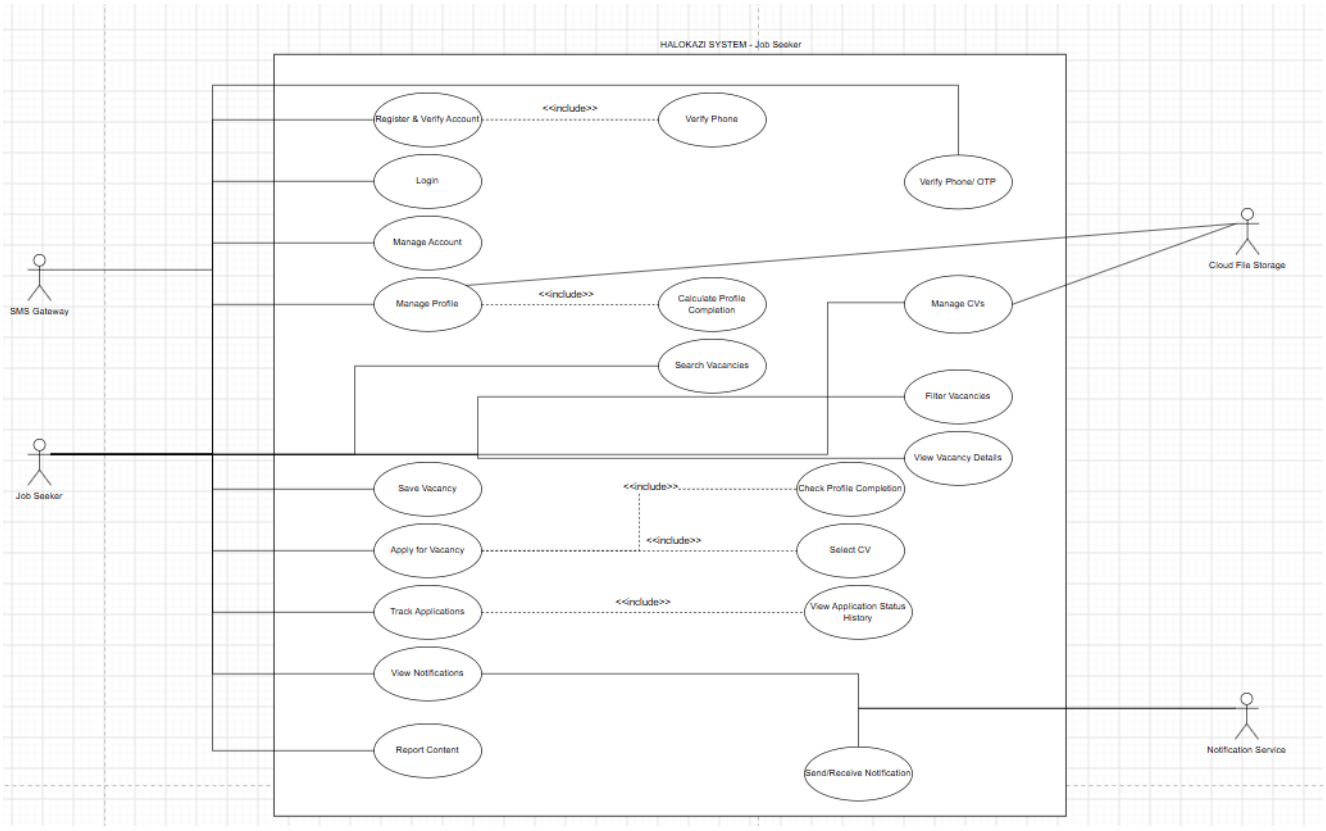
2.2 UC-01 - Overall Halokazi Use Case Diagram

Overall view of Job Seeker, Employer, Admin, Super Admin and secondary services.



2.3 UC-02 - Job Seeker Use Case Diagram

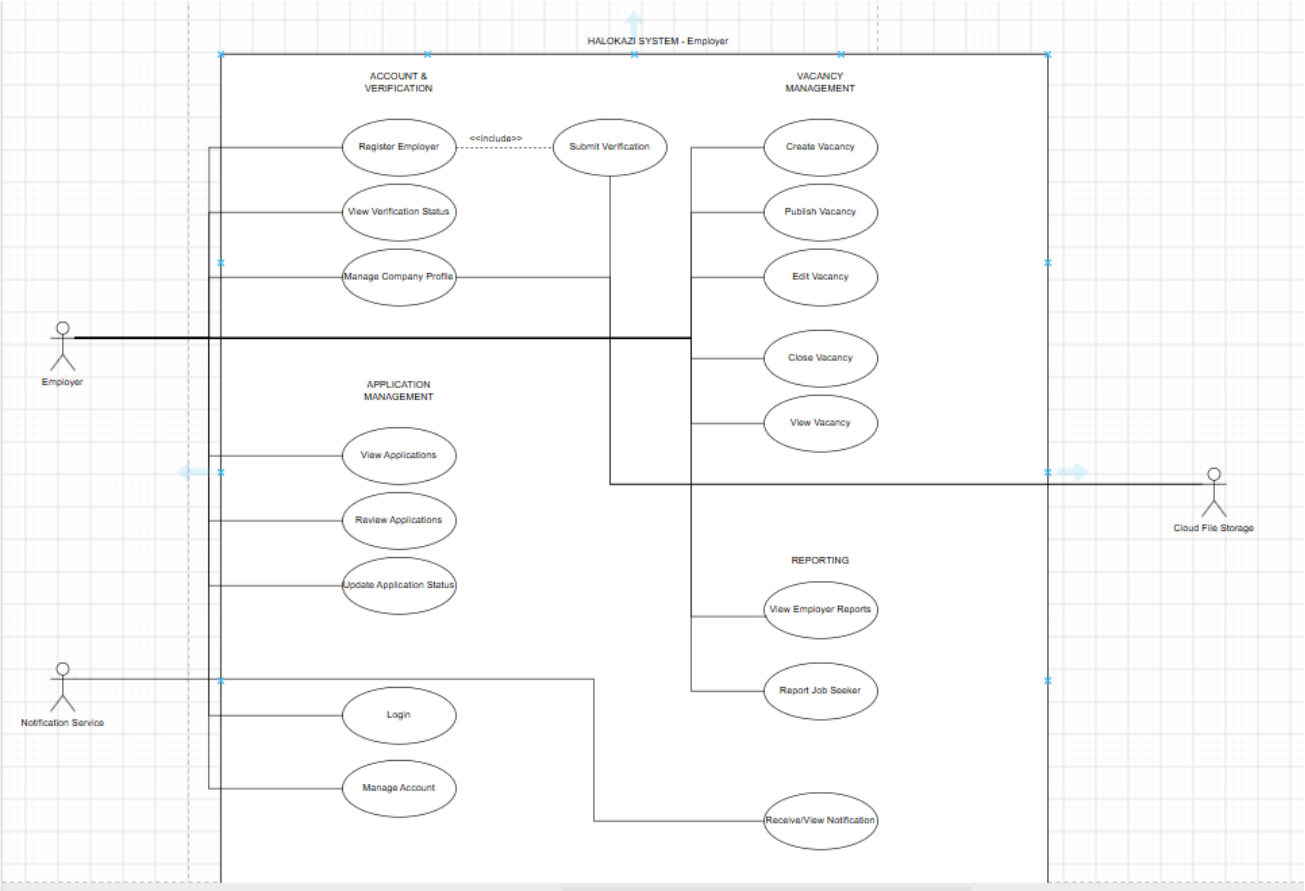
Registration/login, profile management, CV management, vacancy discovery/saving, applications, notifications and reporting.



2.3 UC-02 - Job Seeker Use Case Diagram

2.4 UC-03 - Employer Use Case Diagram

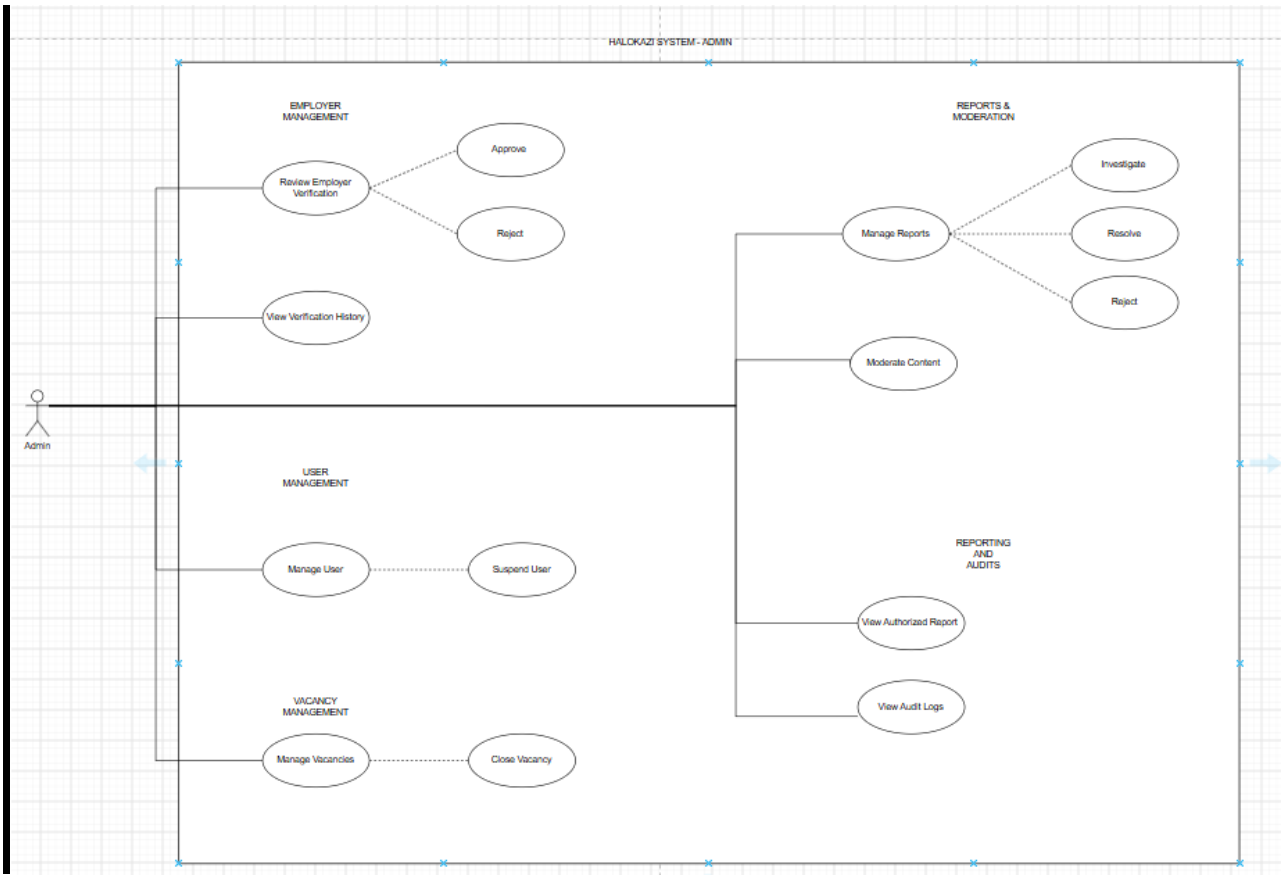
Employer registration/verification, company profile, vacancy management, application review/status, reports and reporting Job Seekers.



2.4 UC-03 - Employer Use Case Diagram

2.5 UC-04 - Admin Use Case Diagram

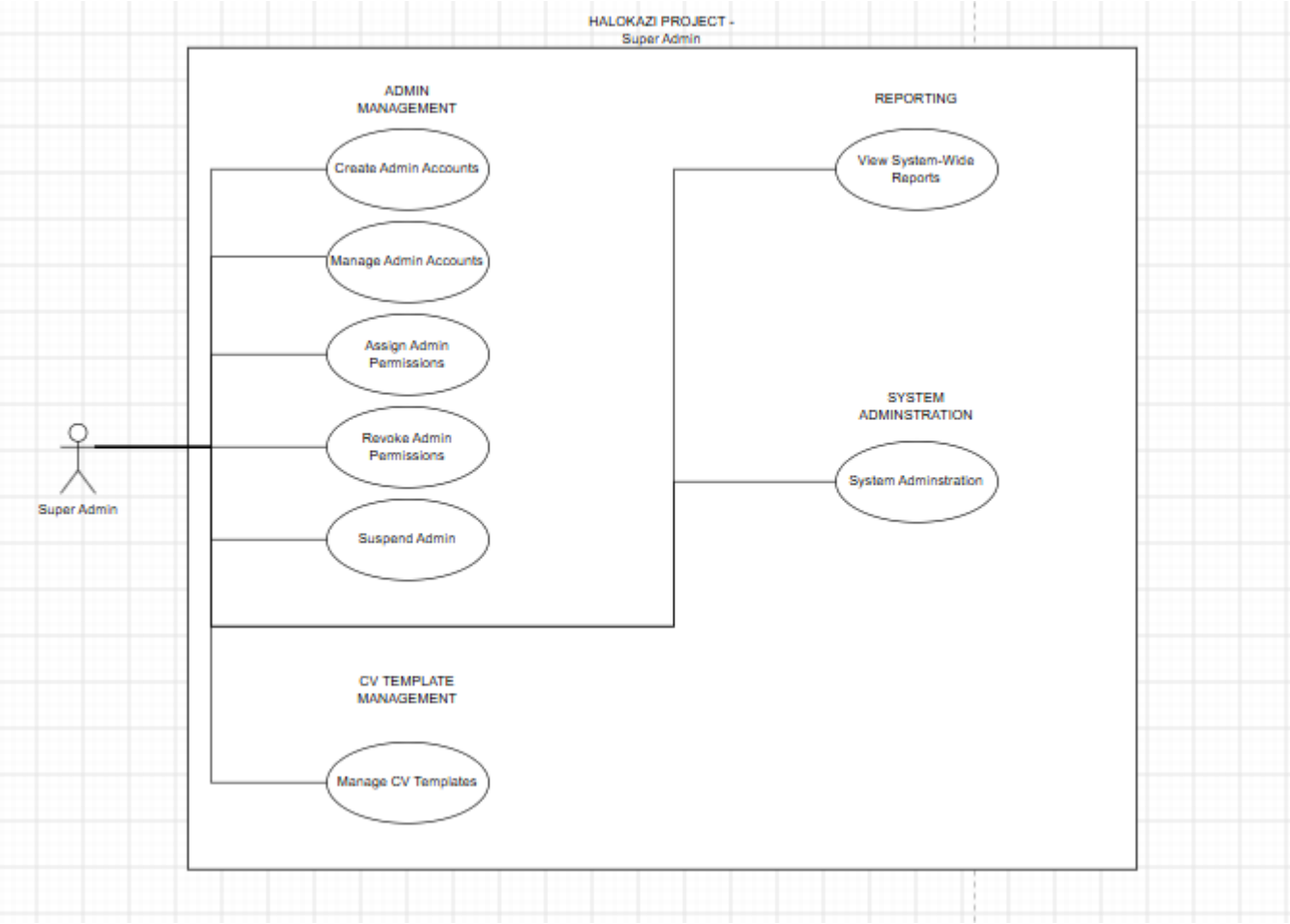
Employer verification, user management, vacancy management, reports, moderation and authorized audit/report access.



2.5 UC-04 — Admin Use Case Diagram

2.6 UC-05 - Super Admin Use Case Diagram

Admin account management, permission assignment/revocation, CV template management, system administration and system-wide reports.



2.6 UC-05 - Super Admin Use Case Diagram

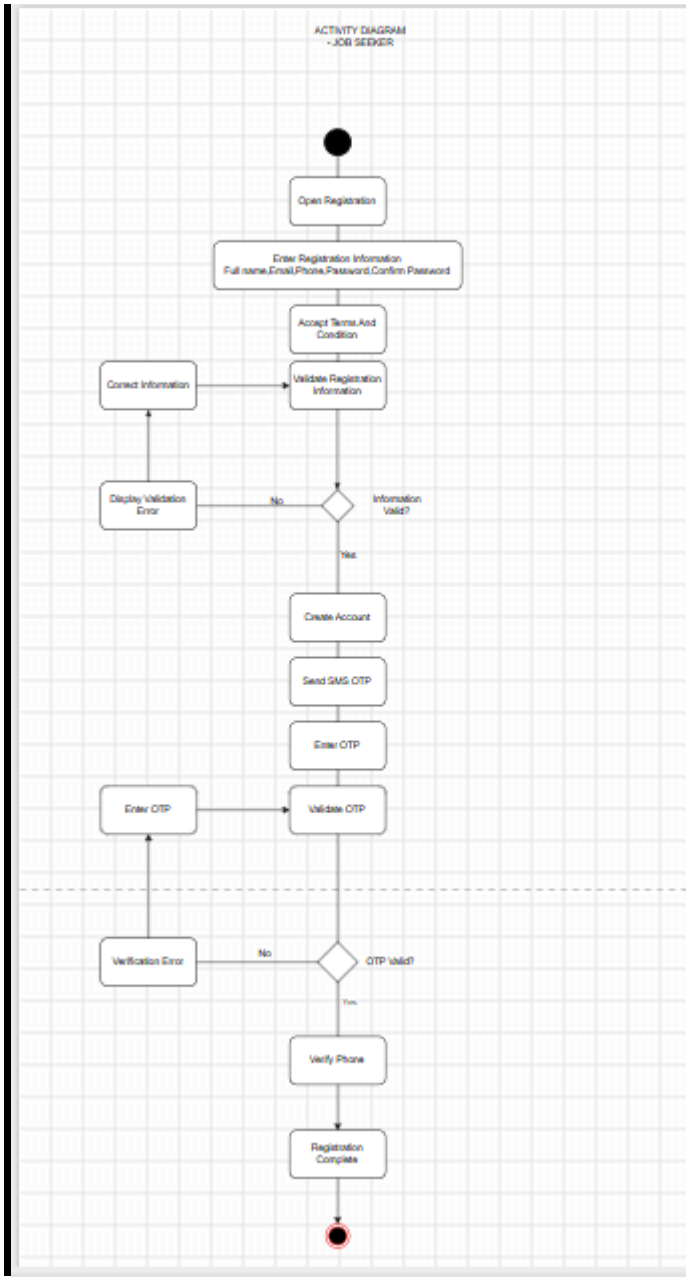
3. Activity Diagrams

The activity diagrams model the sequence of actions, validations, decisions and outcomes for the main operational workflows.

3.1 AD-01 - Job Seeker Registration, Verification & CV Builder

Start → Register → Enter Full Name, Email, Phone, Password and Confirm Password → Accept Terms & Privacy → Validate Details → Valid? If No, correct details and validate again. If Yes, create account → send OTP → enter OTP → OTP Valid? If No, enter OTP again. If Yes, verify phone → login → complete profile → create CV → choose template → fill CV → save CV → End.

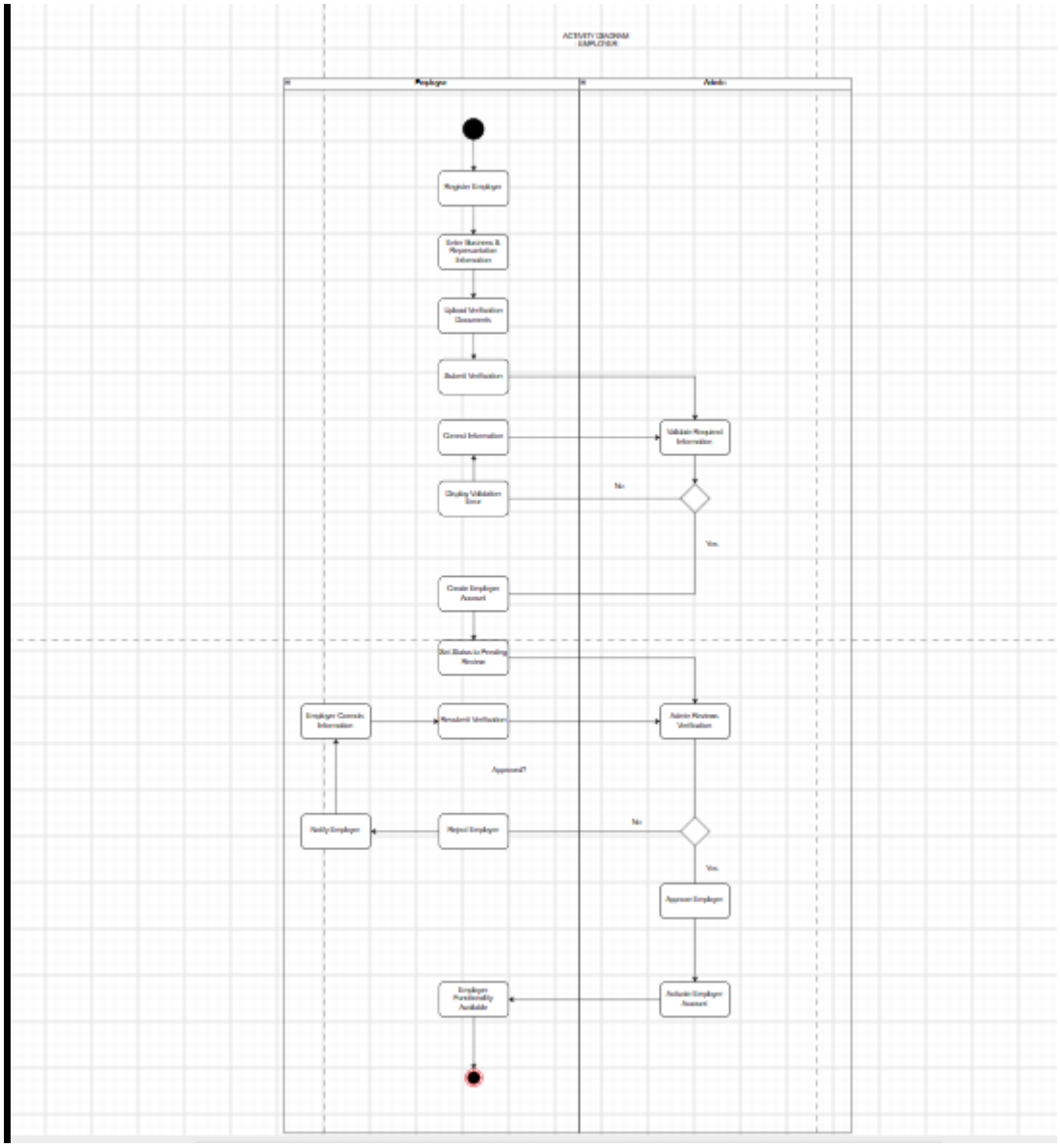
SMS Gateway is used for OTP delivery. Cloud File Storage may support profile/CV attachments.



AD-01 - Job Seeker Registration, Verification & CV Builder

3.2 AD-02 - Employer Registration & Verification

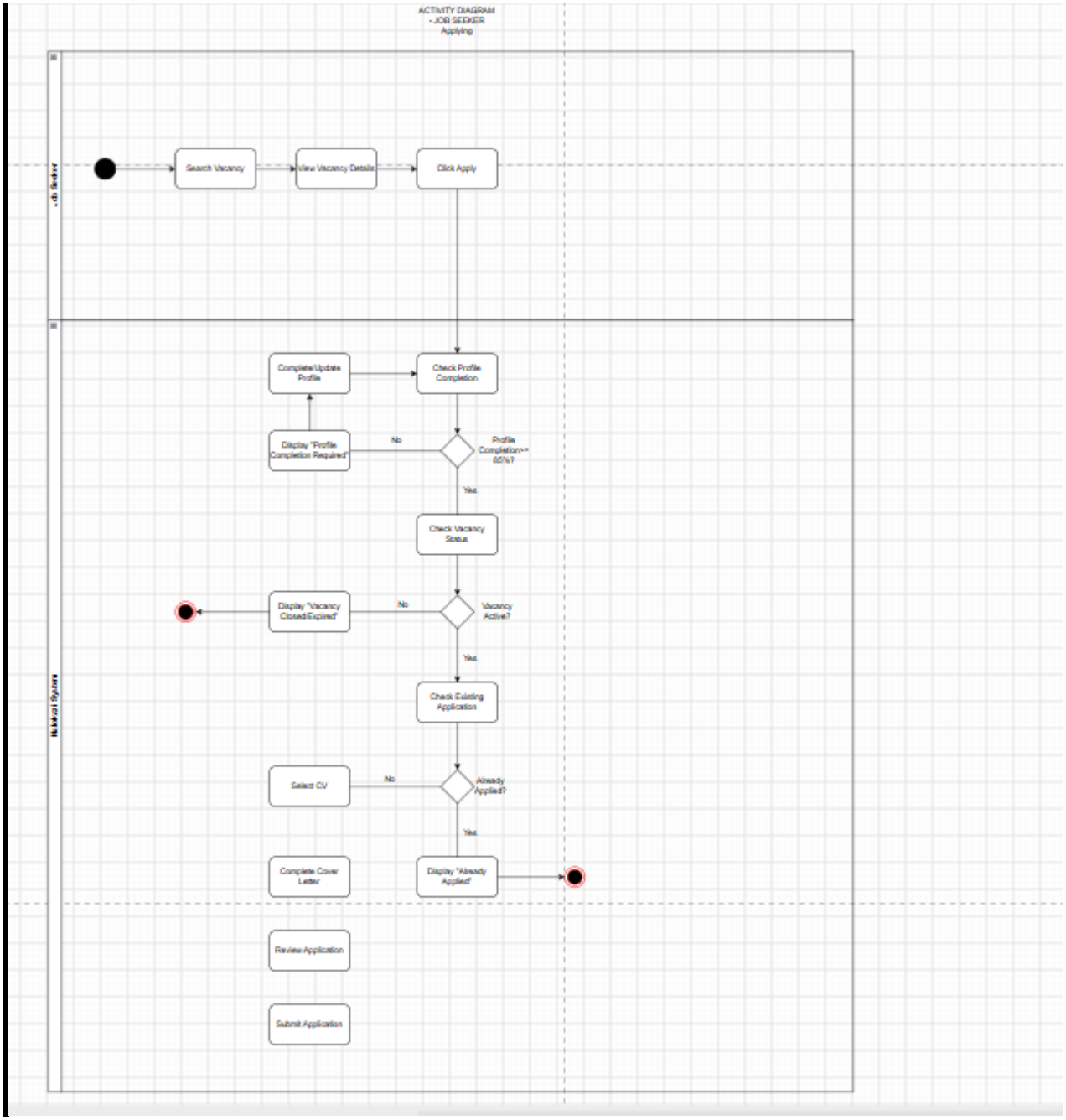
Register Employer → Enter Business Information → Upload Required Documents → Submit Verification → System Validates Information → Valid? If No, correct information and submit again. If Yes, Pending Review → Admin Reviews → Approved? If No, reject and notify, then correct and resubmit. If Yes, approve → activate Employer → Employer functions become available.



AD-02 - Employer Registration & Verification

3.3 AD-03 - Job Seeker Application

Search Vacancy → View Vacancy Details → Apply → Check Profile Completion → Profile ≥85%? If No, complete profile and check again. If Yes, check vacancy active → Active? If No, display closed/expired and end. If Yes, check existing application → Already Applied? If Yes, display already applied and end. If No, select CV → submit certificates → write cover letter → confirm/validate job requirements → review → submit → validate → if invalid, correct/resubmit; if valid, create application → store CV snapshot → Status = Submitted → notification → End.

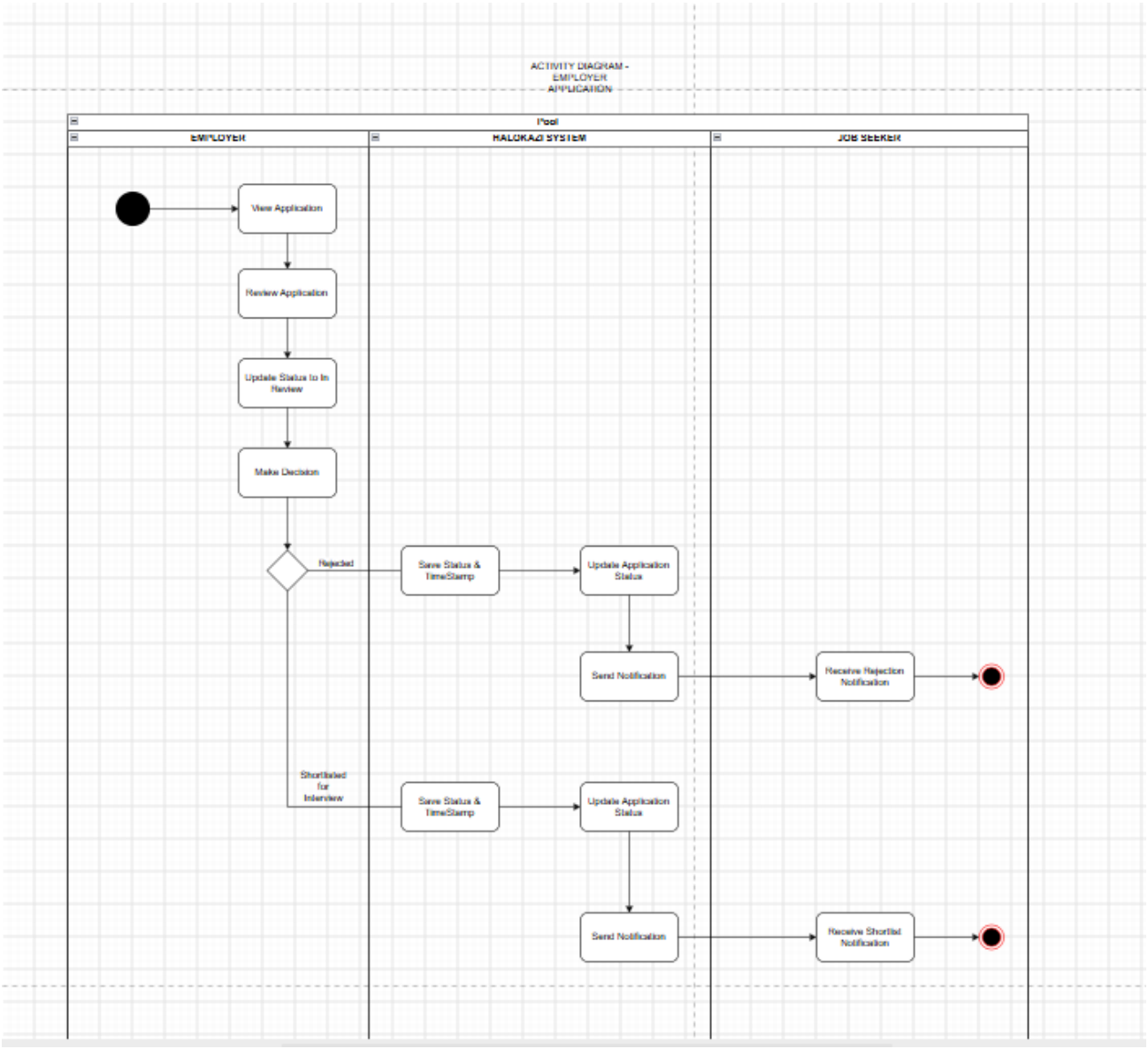


AD-03 — Job Seeker Application

3.4 AD-04 - Employer Application Review & Status Update

Uses three swimlanes: Employer, Halokazi System and Job Seeker. Employer views and reviews an application, changes it to In Review and makes a decision. Rejected → system saves status/timestamp → sends rejection notification → Job Seeker receives notification → End. Shortlisted → system saves status/timestamp → sends shortlist notification → Job Seeker receives notification → End.

Application status: Submitted → In Review → Rejected OR Shortlisted for Interview.

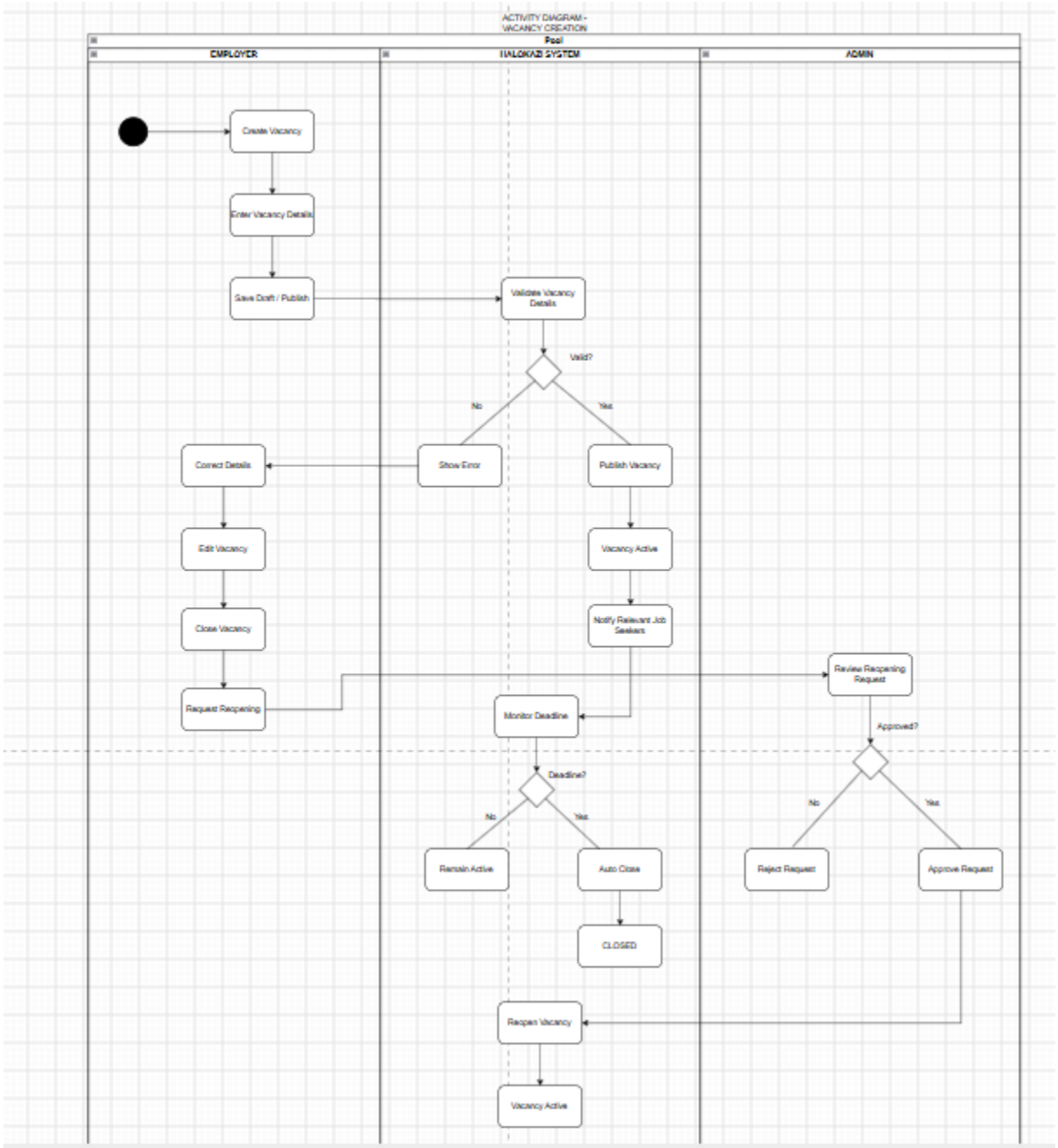


AD-04 - Employer Application Review & Status Update

3.5 AD-05 - Vacancy Creation & Lifecycle

Uses three swimlanes: Employer, Halokazi System and Admin. Employer creates vacancy, enters details and saves draft/publishes. System validates; valid vacancy is published, set Active, made available and relevant Job Seekers notified. System monitors deadline and automatically closes the vacancy when reached.

Employer may edit or manually close a vacancy. A manually closed vacancy requires an Employer reopening request and authorized Admin review. Approval causes the system to reopen and set Active; rejection ends the request.



AD-05 - Vacancy Creation & Lifecycle

4. Entity Relationship Diagram (ERD)

The ERD represents the conceptual database structure for Halokazi, including the main entities, junction entities, keys and relationships.

4.1 User & Access

- USER
- ADMIN_PERMISSION
- ADMIN_PERMISSION_ASSIGNMENT
- POLICY_ACCEPTANCE
- AUDIT_LOG

4.2 Job Seeker Profile

- JOB_SEEKER_PROFILE
- EDUCATION
- WORK_EXPERIENCE
- SKILL
- PROFILE_SKILL
- CERTIFICATION
- LANGUAGE
- PROFILE_LANGUAGE

4.3 CV Management

- CV
- CV_TEMPLATE

4.4 Employer Management

- EMPLOYER
- EMPLOYER_VERIFICATION

4.5 Vacancy Management

- VACANCY
- SAVED_VACANCY

4.6 Application Management

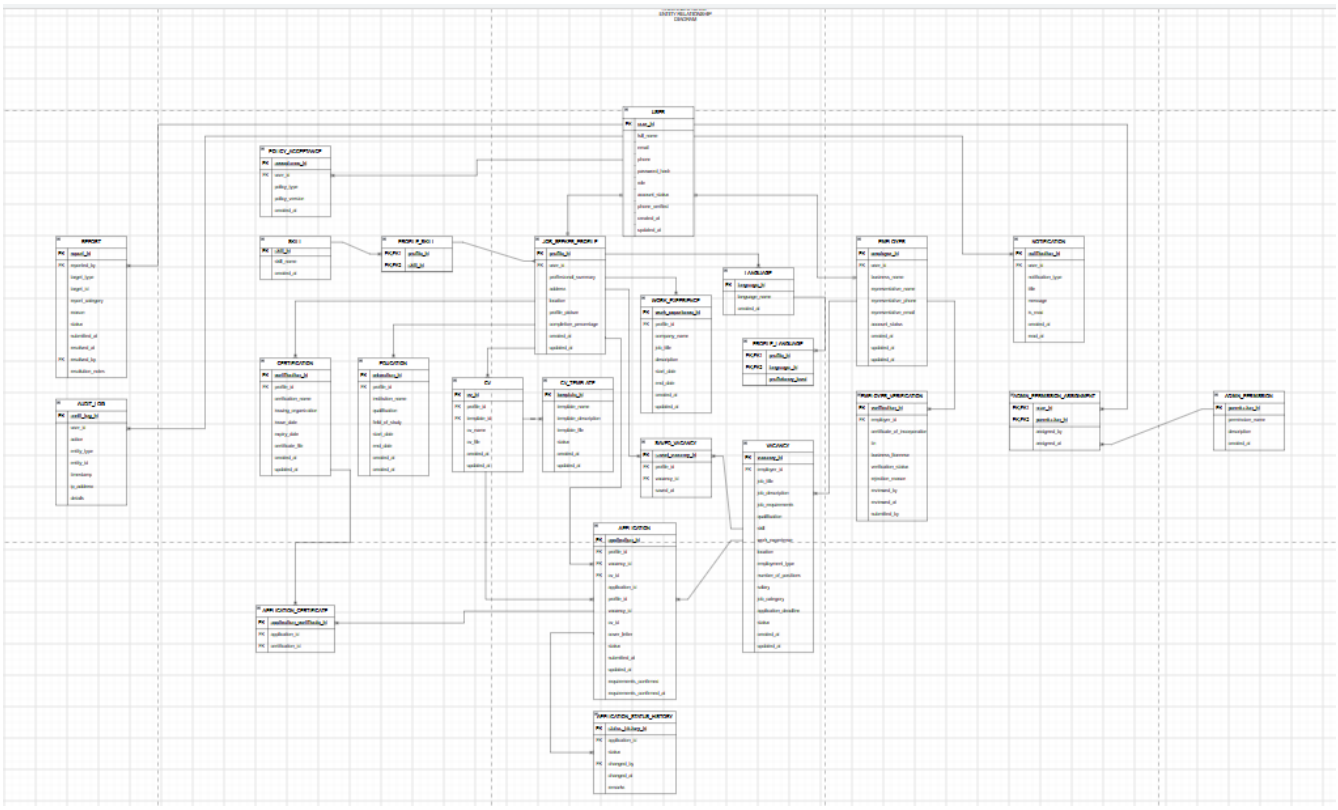
- APPLICATION
- APPLICATION_CERTIFICATE
- APPLICATION_STATUS_HISTORY

4.7 Notifications & Moderation

- NOTIFICATION
- REPORT

4.8 Key Relationships

- USER 1:1 JOB_SEEKER_PROFILE
- USER 1:1 EMPLOYER
- JOB_SEEKER_PROFILE 1:M EDUCATION
- JOB_SEEKER_PROFILE 1:M WORK_EXPERIENCE
- JOB_SEEKER_PROFILE M:N SKILL through PROFILE_SKILL
- JOB_SEEKER_PROFILE 1:M CERTIFICATION
- JOB_SEEKER_PROFILE M:N LANGUAGE through PROFILE_LANGUAGE
- JOB_SEEKER_PROFILE 1:M CV; CV_TEMPLATE 1:M CV
- EMPLOYER 1:M EMPLOYER_VERIFICATION
- EMPLOYER 1:M VACANCY
- JOB_SEEKER_PROFILE 1:M APPLICATION
- VACANCY 1:M APPLICATION
- CV 1:M APPLICATION
- APPLICATION 1:M APPLICATION_STATUS_HISTORY
- APPLICATION M:N CERTIFICATION through APPLICATION_CERTIFICATE
- JOB_SEEKER_PROFILE M:N VACANCY through SAVED_VACANCY
- USER 1:M NOTIFICATION, REPORT and AUDIT_LOG
- USER M:N ADMIN_PERMISSION through ADMIN_PERMISSION_ASSIGNMENT
- USER 1:M POLICY_ACCEPTANCE



ERD - Entity Relationship Diagram

4.9 Database Design Checks

- Remove duplicated application_id, profile_id, vacancy_id and cv_id entries if repeated in APPLICATION.
- Standardize field names and spelling before migrations.
- Confirm submitted_at/submitted_by requirements for EMPLOYER_VERIFICATION.
- Define exact foreign-key targets for reviewed_by, resolved_by, assigned_by and changed_by.
- Keep REPORT.target_id controlled if polymorphic reporting is deliberately supported; otherwise use explicit target relationships.
- Confirm unique constraints for USER email/phone, one Employer account per company and duplicate active applications.
- Preserve the application CV snapshot independently from the source CV.
- Confirm application and vacancy status/lifecycle constraints before migrations.

5. Data Flow Diagrams (DFD)

The DFD models how information moves between Halokazi, external actors/services, internal processes and logical data stores.

5.1 DFD Level 0 — Halokazi Context Diagram

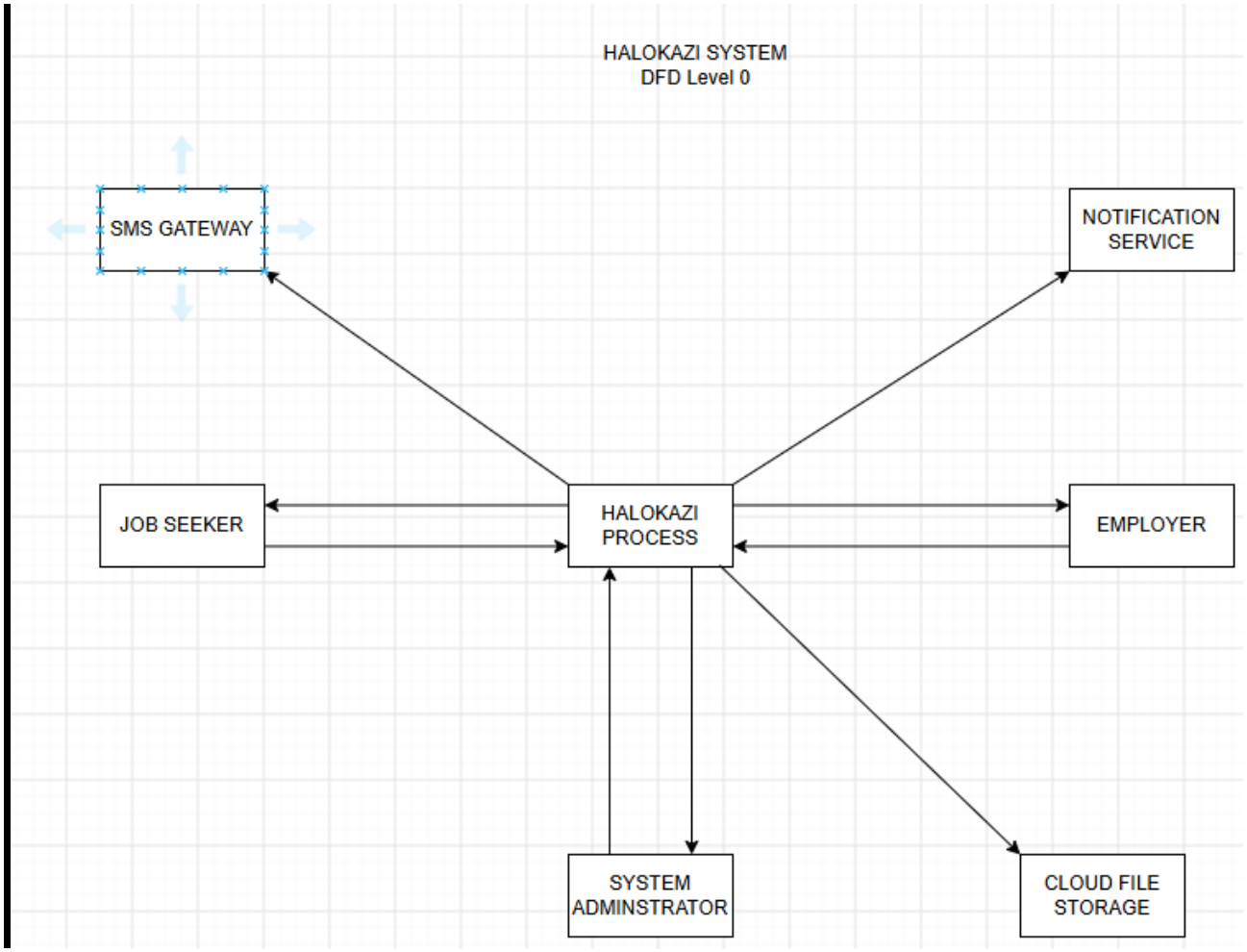
The Level 0 model treats Halokazi as one main system process and shows the major external entities and information exchanged.

External Entities

- Job Seeker
- Employer
- System Administrator
- SMS Gateway
- Notification Service
- Cloud File Storage

Main Data Flows

- Job Seeker → System: registration, profile, CV and application data.
- System → Job Seeker: vacancy information, application status and notifications.
- Employer → System: registration, verification documents, vacancy data and application decisions.
- System → Employer: verification status, applications and applicant information.
- Admin → System: verification decisions and management actions.
- System → Admin: verification data, reports and system information.
- System → SMS Gateway: OTP/SMS data; Gateway → System: delivery result.
- System → Notification Service: notification data; Service → System: delivery result.
- System → Cloud File Storage: CV/profile/employer documents; Storage → System: stored files/retrieval.



DFD Level 0 - Halokazi Context Diagram

5.2 DFD Level 1 — Major Processes and Data Stores

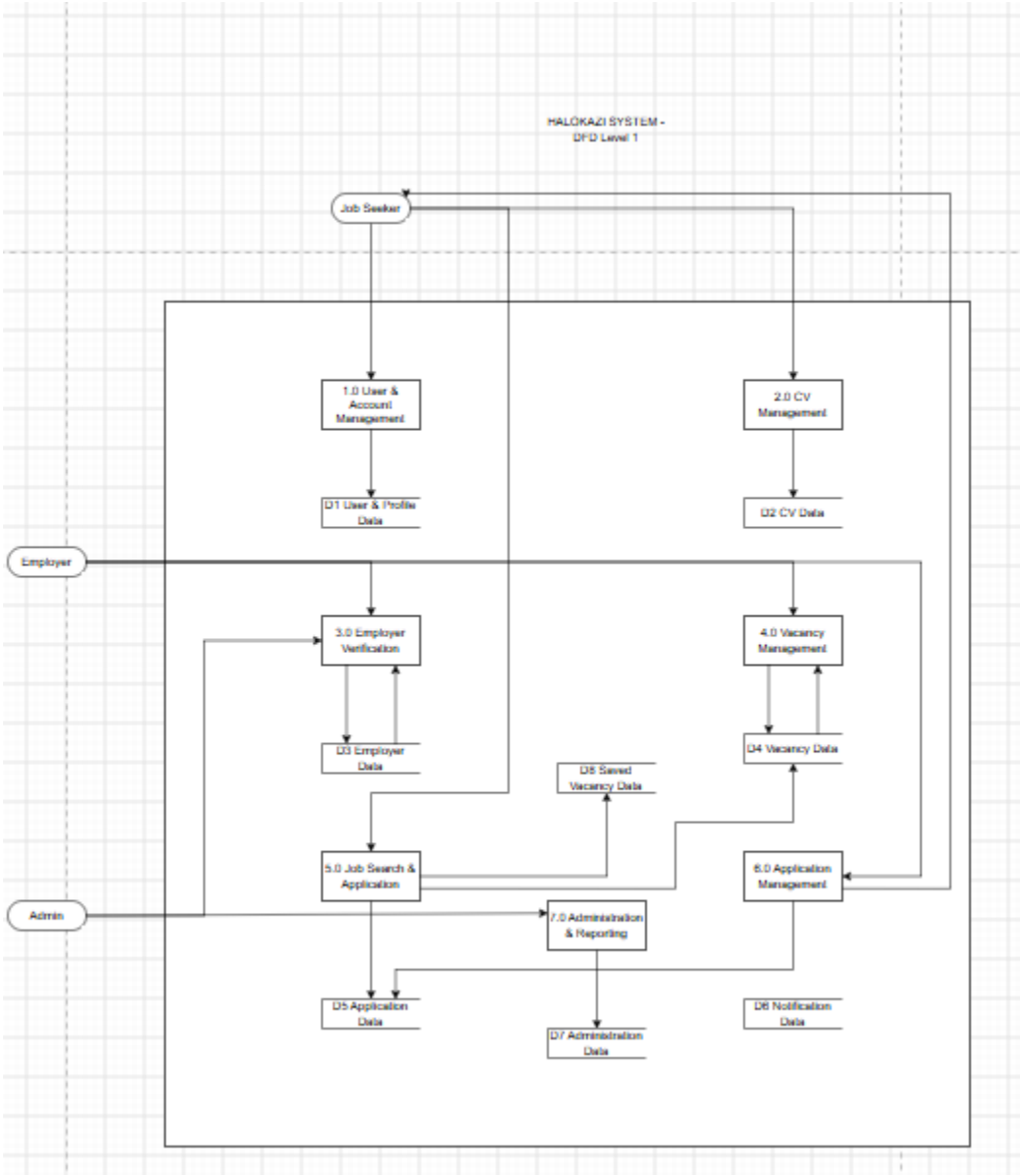
The Level 1 model decomposes Halokazi into major logical processes and data stores. External entities remain outside the system representation, while processes and logical data stores belong inside it.

Major Processes

- 1.0 User & Account Management
- 2.0 CV Management
- 3.0 Employer Verification
- 4.0 Vacancy Management
- 5.0 Job Search & Application
- 6.0 Application Management
- 7.0 Administration & Reporting

Logical Data Stores

- D1 User & Profile Data
- D2 CV Data
- D3 Employer Data
- D4 Vacancy Data
- D5 Application Data
- D6 Notification Data
- D7 Administration Data
- D8 Saved Vacancy Data



DFD Level 1 - Major Processes, Data Stores and Data Flows

6. Modeling Handoff to Database and Design

- Use Case Diagrams identify actors and functions the system must support.
- Activity Diagrams describe sequence, decisions, validations and outcomes.
- DFD Level 0 defines the system context and external information exchanges.
- DFD Level 1 decomposes the system into major processes, data stores and data flows.
- ERD translates the information handled by those processes into entities and relationships for database design.

The ERD provides the conceptual database basis, while the use cases, activities and DFDs provide the functional and data-flow context for detailed database, API and system design.

7. Final Screenshot Compilation Order

- 1. UC-01 Overall Use Case
- 2. UC-02 Job Seeker
- 3. UC-03 Employer
- 4. UC-04 Admin
- 5. UC-05 Super Admin
- 6. AD-01 Job Seeker Registration, Verification & CV Builder
- 7. AD-02 Employer Registration & Verification
- 8. AD-03 Job Seeker Application
- 9. AD-04 Employer Application Review & Status Update
- 10. AD-05 Vacancy Creation & Lifecycle
- 11. ERD
- 12. DFD Level 0
- 13. DFD Level 1

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